

Implementation and Performance Evaluation of an NFT Marketplace on the Sui Blockchain

Chapter 1: Introduction

1.1 Background

Blockchain technology has transformed the way digital assets are created, managed, and exchanged by enabling decentralized ownership and transparent transaction processing. Among the most significant applications of blockchain technology are **Non-Fungible Tokens (NFTs)**, which provide verifiable ownership of unique digital assets such as digital artwork, collectibles, gaming assets, and intellectual property. Unlike cryptocurrencies, NFTs are non-interchangeable because each token possesses unique metadata and ownership information recorded permanently on the blockchain.

As NFT ecosystems continue to grow, marketplaces have become an essential component for enabling users to mint, list, buy, and trade digital assets without relying on centralized intermediaries. However, conventional blockchain platforms often face challenges such as high transaction latency, limited throughput, network congestion, and increasing transaction costs. These limitations affect user experience and reduce the practicality of decentralized NFT marketplaces under large-scale adoption.

To overcome these challenges, newer blockchain platforms have introduced alternative execution models that improve scalability while maintaining decentralization and security. One such platform is **Sui**, a Layer-1 blockchain developed using the Move programming language. Sui introduces an object-centric architecture that differs significantly from traditional account-based blockchains by treating digital assets as independently owned objects. This design enables efficient ownership management and allows independent transactions to execute in parallel, thereby improving scalability and reducing unnecessary consensus overhead.

The NFT marketplace developed in this project leverages these architectural features to implement the complete lifecycle of NFT operations, including asset creation, marketplace listing, purchasing, and listing cancellation. The implementation also incorporates a loyalty module to demonstrate how additional decentralized services can be integrated into the marketplace.

1.2 Motivation

The selection of the Sui blockchain for this implementation was motivated by several architectural advantages that directly support NFT applications.

Unlike traditional blockchain systems where all transactions compete for shared global state, Sui distinguishes between owned objects and shared objects. Transactions involving independently owned objects can be executed without global consensus, enabling significantly greater parallelism compared to conventional blockchain architectures. Since NFTs naturally represent individually owned assets, they are particularly well suited to Sui's execution model.

Furthermore, Sui utilizes the Move programming language, which provides resource-oriented programming semantics that prevent common smart contract vulnerabilities such as accidental asset duplication or unintended deletion. These language-level guarantees simplify secure smart contract development while improving overall reliability.

The combination of object ownership, parallel transaction execution, and resource safety makes Sui an attractive platform for implementing decentralized NFT marketplaces capable of supporting future large-scale adoption.

1.3 Objectives

The primary objective of this report is to present the implementation, deployment, and performance evaluation of an NFT marketplace developed on the Sui blockchain.

The specific objectives are as follows:

- Design and implement an NFT marketplace using the Move programming language.
- Develop smart contract modules for NFT management, marketplace operations, and user loyalty rewards.
- Deploy the smart contract package successfully on the Sui Testnet.
- Validate the correctness of all marketplace operations through functional testing.
- Evaluate marketplace performance using latency, throughput, and gas consumption measurements.
- Analyse the strengths and limitations of Sui's object-oriented execution model for NFT marketplace applications.

1.4 Scope of the Report

This report focuses exclusively on the implementation of the NFT marketplace on the **Sui blockchain**. The discussion includes the architecture of the developed smart contracts, deployment procedure, functional validation, and experimental performance benchmarking conducted on the public Sui Testnet.

The report does not include a detailed security audit or comparative analysis with other blockchain platforms, as these topics are addressed separately in the project's Security Audit Report and Cross-Platform Performance Evaluation Report.

Chapter 2: Overview of the Sui Blockchain

2.1 Introduction to Sui

Sui is a next-generation Layer-1 blockchain designed to address the scalability limitations of traditional distributed ledger systems while maintaining decentralization and security. Developed by Mysten Labs, Sui introduces a fundamentally different execution model in which digital assets are represented as programmable objects rather than balances maintained within user accounts. This object-centric architecture enables more efficient state management and allows transactions involving independent objects to execute in parallel, significantly improving network throughput.

Unlike conventional account-based blockchains, where every transaction potentially modifies a shared global state, Sui distinguishes between **owned objects** and **shared objects**. Transactions involving owned objects can often be executed independently without requiring global consensus, reducing execution latency and increasing scalability. Shared objects, on the other hand, are synchronized through consensus to maintain consistency across all network validators. This hybrid execution model allows Sui to balance performance with correctness, making it particularly suitable for decentralized applications that manage large numbers of digital assets.

The Sui blockchain is implemented using the **Move programming language**, originally developed for secure digital asset management. Move introduces resource-oriented programming concepts that ensure digital assets cannot be accidentally duplicated or destroyed, thereby providing strong guarantees regarding ownership and asset integrity.

These architectural characteristics make Sui an attractive platform for implementing NFT marketplaces, where each NFT naturally exists as an independent digital object with unique ownership.

2.2 Object-Centric Architecture

The defining feature of the Sui blockchain is its object-centric execution model. Every asset stored on the blockchain is represented as an object possessing a globally unique identifier, ownership information, version history, and associated metadata. Instead of updating account balances, transactions directly manipulate these objects according to the rules defined by the deployed smart contracts.

Objects in Sui are broadly classified into two categories:

- **Owned Objects**
- **Shared Objects**

Owned objects belong exclusively to a single user and may be transferred directly between owners. NFTs created within the marketplace belong to this category because each token represents a unique asset owned by an individual account.

Shared objects are accessible by multiple users and therefore require validator consensus whenever they are modified. In the developed marketplace, the marketplace registry that stores active NFT listings is implemented as a shared object to ensure that all participants observe a consistent marketplace state.

This separation between owned and shared objects allows Sui to process independent NFT transactions concurrently while maintaining consistency for marketplace operations involving shared data structures.

2.3 Move Programming Language

The Sui blockchain utilizes the Move programming language for smart contract development. Move is specifically designed for secure management of digital assets through a resource-oriented programming model.

Unlike conventional programming languages where variables can often be copied or discarded freely, Move treats digital assets as **resources**. Resources possess strict ownership semantics and cannot be duplicated or implicitly destroyed. Every resource must always have exactly one owner unless ownership is intentionally transferred according to the smart contract logic.

For the developed NFT marketplace, Move provides several important advantages:

- NFTs cannot be duplicated accidentally.
- Ownership transfers occur explicitly.
- Unauthorized asset manipulation is prevented.
- Strong compile-time checks reduce programming errors.
- Resource safety improves overall smart contract reliability.

These properties contribute significantly to the secure implementation of marketplace operations.

2.4 Parallel Transaction Execution

Scalability is one of the primary objectives of the Sui blockchain. Traditional blockchain systems execute transactions sequentially because they often compete to modify the same global state. This limits throughput as the number of users increases.

Sui addresses this limitation through object-level parallelism.

When two transactions operate on different owned objects without modifying shared state, they can execute simultaneously on different validators without waiting for one another. For example, two users minting independent NFTs can perform these operations concurrently because each transaction creates separate objects.

However, transactions involving shared marketplace objects—such as listing or cancelling NFTs—must still undergo consensus to preserve consistency. Consequently, these operations typically exhibit higher execution latency than transactions involving only owned objects.

This execution model enables Sui to achieve significantly greater scalability for applications where the majority of transactions involve independent assets.

2.5 Smart Contract Modules Used in the Project

The NFT marketplace developed in this project consists of three Move modules, each responsible for a specific aspect of marketplace functionality.

NFT Module

The NFT module manages the creation of unique NFT objects. During minting, each NFT receives a unique identifier together with associated metadata such as the asset name, description, and owner information. The module ensures that newly created NFTs are transferred directly to the creator upon successful transaction execution.

Marketplace Module

The marketplace module enables users to list NFTs for sale, purchase listed assets, and cancel active listings. Marketplace information is maintained through shared objects, allowing all participants to access a consistent view of available listings while preserving transaction correctness.

Loyalty Module

To demonstrate extensibility beyond basic marketplace operations, a loyalty module was implemented. This module records reward points earned by users during marketplace activity. Although it does not directly participate in NFT ownership transfer, it illustrates how additional decentralized services can be integrated into the overall marketplace architecture.

Together, these three modules implement the complete lifecycle of NFT creation, trading, and user engagement.

2.6 Why Sui for NFT Marketplaces?

Several architectural features make Sui particularly suitable for decentralized NFT marketplaces.

First, NFTs naturally align with Sui's object-oriented execution model because each asset already represents an independent object with unique ownership. This simplifies ownership management and reduces the complexity of smart contract development.

Second, Move's resource-oriented programming model provides strong safety guarantees that eliminate common asset-management vulnerabilities such as accidental duplication or unintended deletion of NFTs.

Third, Sui's ability to execute independent object transactions in parallel offers the potential for significantly higher throughput than conventional sequential blockchain systems. Although the benchmark presented later in this report measures sequential CLI execution on the public testnet, the underlying architecture is capable of substantially greater performance under parallel workloads.

Finally, the distinction between owned and shared objects allows the marketplace to balance efficiency and consistency. Individual NFT operations benefit from parallel execution, while marketplace listings remain synchronized through consensus, ensuring a reliable trading environment.

Chapter 3: NFT Marketplace Design and Implementation

3.1 System Overview

The NFT marketplace developed in this project provides a decentralized platform where users can create, trade, and manage Non-Fungible Tokens (NFTs) without relying on a centralized authority. The marketplace was implemented using the **Move programming language** on the **Sui blockchain**, leveraging its object-centric architecture to ensure secure ownership management and efficient transaction execution.

The system supports the complete lifecycle of an NFT through four primary marketplace operations:

- NFT Minting
- NFT Listing
- NFT Purchase
- Listing Cancellation

In addition to these core functionalities, a loyalty mechanism was incorporated to demonstrate the extensibility of the platform by rewarding users for marketplace participation.

The marketplace architecture is modular, with individual smart contract modules responsible for specific functionalities. This modular design improves maintainability, simplifies testing, and allows future enhancements without affecting unrelated components.

Figure 3.1 illustrates the overall architecture of the developed marketplace.

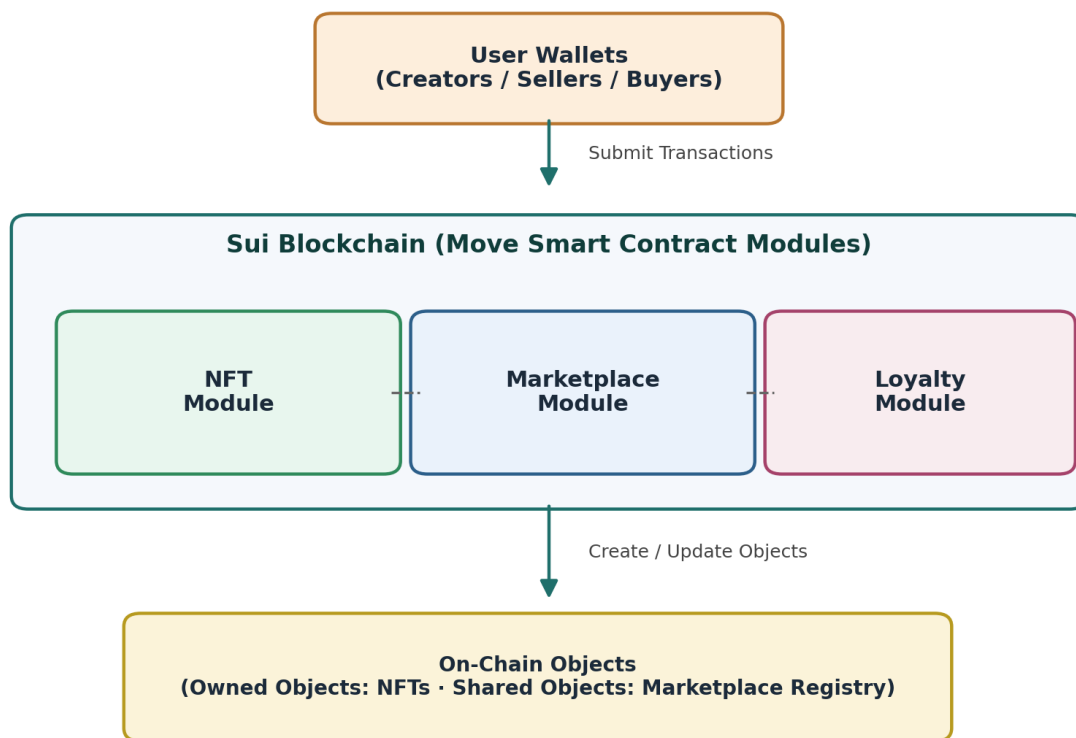


Figure 3.1 Overall Architecture of the NFT Marketplace on Sui

The architecture consists of three primary smart contract modules together with user wallets interacting through the Sui blockchain. Each module performs a distinct role while collectively supporting the complete marketplace workflow.

3.2 Smart Contract Architecture

The marketplace package consists of three Move modules:

- NFT Module
- Marketplace Module
- Loyalty Module

Each module is designed according to the principle of separation of responsibilities.

The interaction among these modules enables users to create digital assets, trade them securely, and receive incentives for marketplace participation.

The relationships among the modules are illustrated in Figure 3.2.

Interaction between Smart Contract Modules

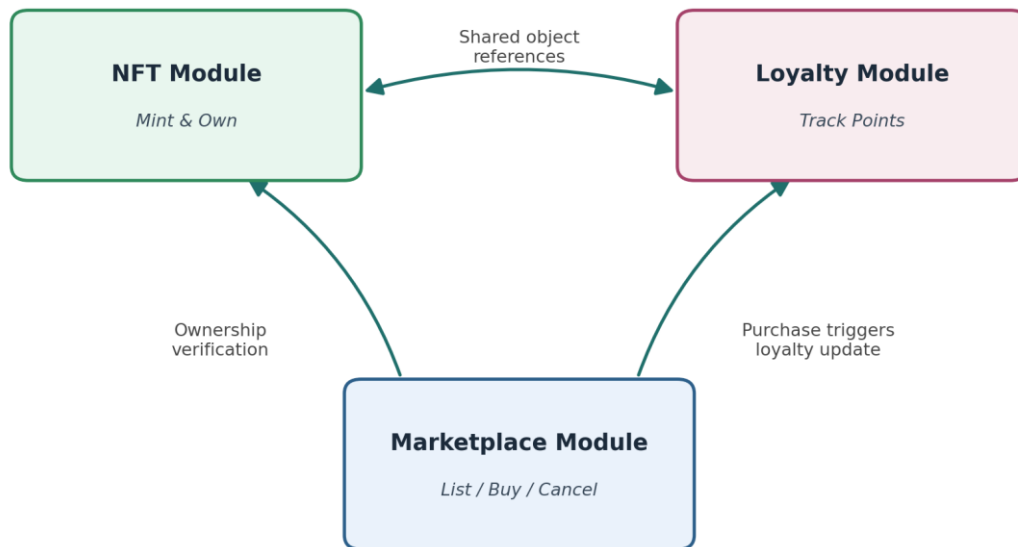


Figure 3.2 Interaction between Smart Contract Modules

3.3 NFT Module

The NFT module is responsible for creating unique digital assets on the blockchain. During the minting process, a new NFT object is generated and assigned to the creator. Each NFT contains metadata describing the digital asset, including its identifier, name, description, and ownership information.

Because Sui treats NFTs as owned objects, every NFT possesses a globally unique object identifier that distinguishes it from all other blockchain assets. Ownership is transferred directly to the creator immediately after successful transaction execution.

The minting process consists of the following steps:

1. User submits a mint transaction.
2. Smart contract creates a new NFT object.
3. Metadata is initialized.
4. Ownership is assigned to the creator.
5. NFT object is transferred to the user's wallet.

The use of Move resources ensures that NFT objects cannot be duplicated or unintentionally destroyed during execution.

3.4 Marketplace Module

The marketplace module manages all trading activities performed within the application.

Its responsibilities include:

- Creating marketplace listings.
- Maintaining active sale records.
- Executing NFT purchases.
- Removing completed or cancelled listings.

Unlike NFT objects, marketplace information is maintained using a shared marketplace object because multiple users may simultaneously access marketplace data.

Whenever a seller lists an NFT, the marketplace module records:

- NFT Identifier
- Seller Address
- Sale Price
- Listing Status

This shared marketplace object enables buyers to retrieve available listings while ensuring that all marketplace participants observe a consistent system state.

Listing Workflow

The listing process proceeds as follows:

1. Seller selects an NFT.
2. Ownership verification is performed.
3. Listing information is stored in the marketplace.
4. Marketplace state is updated.
5. NFT becomes available for purchase.

The marketplace rejects listing requests if ownership verification fails.

Purchase Workflow

The purchase process includes several coordinated operations.

First, the marketplace verifies that the requested NFT is actively listed for sale. The buyer then submits the required payment, after which ownership of the NFT is transferred from the seller to the buyer. Finally, the marketplace removes the completed listing from the active marketplace records.

Because transaction execution on Sui is atomic, either all of these operations complete successfully or none of them are applied. This prevents inconsistent marketplace states and guarantees ownership correctness.

Listing Cancellation

The marketplace also allows sellers to cancel active listings.

Cancellation consists of:

- Seller authentication.
- Marketplace verification.
- Listing removal.
- Marketplace update.

Since ownership never leaves the seller during listing, cancelling a listing simply removes the NFT from the marketplace without requiring ownership transfer.

3.5 Loyalty Module

To demonstrate the extensibility of the marketplace beyond basic NFT trading, a loyalty module was developed as an additional smart contract component.

The loyalty module records user reward points earned through marketplace participation. Reward points may be accumulated following successful marketplace interactions, encouraging continued engagement with the platform.

Although the current implementation focuses on recording reward information, the module establishes a foundation for future enhancements such as:

- Discount mechanisms.
- Membership levels.
- Reward redemption.

- Marketplace incentives.

Separating the loyalty logic from the marketplace module improves modularity and simplifies future expansion.

3.6 Marketplace Workflow

The overall workflow of the NFT marketplace follows the complete lifecycle of an NFT from creation to successful sale.

The sequence of operations is illustrated below.

1. User creates a wallet.
2. User mints an NFT.
3. NFT is stored in the user's wallet.
4. User lists the NFT on the marketplace.
5. Marketplace records the listing.
6. Buyer purchases the NFT.
7. Ownership transfers to the buyer.
8. Marketplace removes the completed listing.
9. Loyalty information is updated.

Complete NFT Marketplace Workflow

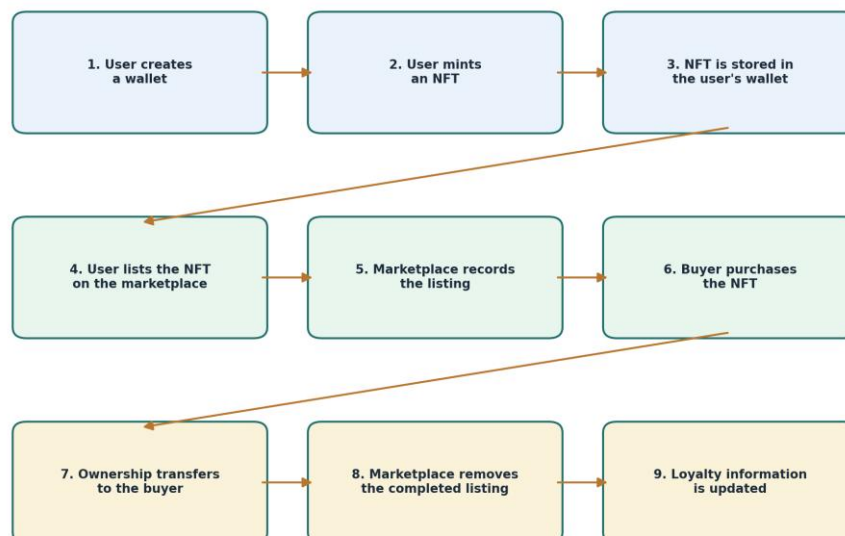


Figure 3.3 Complete NFT Marketplace Workflow

This workflow demonstrates how the individual smart contract modules cooperate to execute marketplace transactions while maintaining ownership integrity throughout the NFT lifecycle.

3.7 Design Advantages

Several architectural decisions contribute to the efficiency and maintainability of the developed marketplace.

The modular smart contract structure separates NFT management, marketplace functionality, and loyalty management into independent components. This improves readability and allows individual modules to evolve without affecting the overall system.

Sui's object-oriented execution model naturally aligns with NFT ownership, simplifying ownership management while enabling secure transfer of assets between users.

The use of shared marketplace objects ensures that listing information remains synchronized across all participants, while Move's resource-oriented programming model guarantees that NFTs cannot be duplicated or accidentally destroyed.

Together, these design choices provide a secure, extensible, and scalable architecture suitable for decentralized NFT marketplace applications.

Chapter 4: Testnet Deployment and Functional Validation

4.1 Introduction

Following the successful implementation of the NFT marketplace smart contracts, the next phase involved deploying the package on the **Sui Testnet** and validating the execution of all marketplace operations. Public testnet deployment provides a realistic environment for evaluating smart contract behaviour under actual blockchain conditions without requiring a production deployment.

The primary objectives of this phase were to verify that the smart contract package compiled successfully, could be published to the Sui blockchain, and correctly executed all core marketplace operations. Functional validation also ensured that ownership transfers, marketplace state updates, and transaction execution behaved as intended.

4.2 Experimental Environment

The implementation and deployment were performed using the official Sui development tools. The smart contract package was written in the Move programming language and deployed using the Sui Command Line Interface (CLI).

Table 4.1 summarizes the experimental environment used during deployment.

Table 4.1 Development Environment

Component	Description
Blockchain Platform	Sui Testnet
Smart Contract Language	Move
Development Framework	Sui CLI
Network	Public Testnet
Package Type	Move Package
Deployment Method	CLI Package Publish

The public testnet was selected because it closely resembles the behaviour of the main network while allowing unrestricted testing and experimentation.

4.3 Smart Contract Compilation

Before deployment, the marketplace package was compiled to verify the correctness of the implemented Move modules.

The compilation process validates:

- syntax correctness,
- type safety,
- module dependencies,
- resource ownership rules,
- package consistency.

The compilation was performed using the following command:

```
sui move build
```

The compilation completed successfully without any errors or warnings, indicating that all three modules—NFT, Marketplace, and Loyalty—satisfied the requirements of the Move compiler.

Successful compilation also confirms that the implemented resource management rules comply with Move's ownership model and that the package is ready for deployment.

4.4 Unit Testing

Following successful compilation, the implemented smart contracts were validated using Move's built-in testing framework.

Testing was performed using:

```
sui move test
```

All implemented unit tests executed successfully.

The successful execution of the test suite confirmed:

- Correct module initialization.
- Proper object creation.
- Successful ownership assignment.
- Correct transaction execution.
- Expected marketplace behaviour.

Passing all unit tests provided confidence that the marketplace logic behaved correctly before deployment to the public testnet.

4.5 Package Deployment

After successful compilation and testing, the Move package was published to the Sui Testnet.

The deployment process involved:

1. Building the Move package.
2. Publishing the package.
3. Registering the package on the blockchain.
4. Receiving the Package ID.
5. Initializing marketplace objects.

Deployment completed successfully without transaction failures.

Once published, the marketplace package became permanently accessible on the Sui Testnet, allowing interaction through subsequent marketplace transactions.

The successful deployment demonstrated that the developed smart contracts were compatible with the Sui execution environment and could operate correctly on a decentralized blockchain network.

4.6 Functional Validation

Following deployment, the complete NFT marketplace workflow was evaluated through real blockchain transactions.

The following operations were executed successfully:

- NFT Minting
- NFT Listing
- NFT Purchase
- Listing Cancellation

Each transaction was confirmed on the Sui Testnet, indicating successful execution by the validator network.

The execution of these operations verified that the marketplace functions correctly under real network conditions.

Table 4.2 summarizes the functional validation results.

Table 4.2 Functional Validation Results

Marketplace Operation Status

Mint NFT	Successful
List NFT	Successful
Buy NFT	Successful
Cancel Listing	Successful

The successful execution of all four operations demonstrates that the developed marketplace supports the complete NFT lifecycle from asset creation to ownership transfer.

4.7 Ownership Verification

An essential objective of functional validation was to verify that NFT ownership remained consistent throughout the marketplace workflow.

Ownership validation confirmed that:

- Newly minted NFTs were assigned to the creator.
- Listed NFTs remained associated with the seller until purchase.
- Ownership transferred correctly after successful purchases.
- Cancelled listings remained under seller ownership.
- Marketplace records remained synchronized with NFT ownership.

No inconsistencies in ownership information were observed during testing.

These results confirm that the marketplace correctly preserves ownership integrity throughout every transaction.

4.8 Marketplace State Validation

The marketplace maintains listing information using a shared object that records active NFTs available for sale.

During testing, the following state transitions were verified:

- New listings were successfully added to the marketplace.

- Purchased NFTs were automatically removed from active listings.
- Cancelled listings were deleted from marketplace records.
- Marketplace data remained consistent after every transaction.

No duplicate listings or stale marketplace records were observed during the evaluation.

4.9 Discussion

The deployment and validation results demonstrate that the implemented NFT marketplace functions correctly on the Sui Testnet. Successful compilation, unit testing, package publication, and execution of live transactions confirm the correctness of the implemented smart contracts and the reliability of the overall marketplace workflow.

The evaluation also highlights the practical advantages of Sui's object-centric architecture. NFT ownership was managed through independently owned objects, while marketplace listings were maintained using shared objects to ensure consistency among all users. This combination enabled secure execution of marketplace operations without compromising ownership integrity.

Although testing was conducted on the public testnet, which may introduce additional network latency compared to optimized production environments, the observed behaviour confirms that the implementation is functionally complete and suitable for performance evaluation.

Chapter 5: Performance Benchmarking

5.1 Introduction

Performance evaluation is a critical aspect of blockchain-based application development, as it provides insights into the practical efficiency of smart contract execution under real network conditions. For decentralized NFT marketplaces, performance directly affects user experience, transaction responsiveness, and the scalability of the platform. Metrics such as transaction latency, throughput, and gas consumption help determine whether the underlying blockchain architecture is capable of supporting real-world workloads.

In this project, the developed NFT marketplace was benchmarked on the **Sui Testnet** after successful deployment and functional validation. The experiments focused on measuring the execution performance of the four primary marketplace operations: NFT minting, listing, purchasing, and listing cancellation. Additionally, a throughput experiment was conducted by executing a batch of minting transactions sequentially, and gas consumption data was collected from transaction execution outputs.

The objective of this benchmarking exercise was not to determine the maximum theoretical performance of the Sui blockchain, but rather to establish a practical baseline for the implemented marketplace under realistic testnet conditions.

5.2 Experimental Methodology

The benchmarking experiments were conducted using the official Sui CLI on the public Sui Testnet. Each marketplace operation was executed individually, and the total execution time was recorded from transaction submission until confirmation. Since the experiments relied on live network interactions, the measured latency includes blockchain processing time, validator communication, consensus delays where applicable, and network latency.

The benchmark focused on the following marketplace operations:

- **mint_nft** – Creates a new NFT object and assigns ownership to the creator.
- **list_nft** – Registers an NFT for sale within the marketplace.
- **buy_nft** – Transfers ownership of a listed NFT from the seller to the buyer.
- **cancel_listing** – Removes an active NFT listing from the marketplace.

To evaluate throughput, a batch of five NFT minting transactions was executed sequentially through the CLI. The total execution time was measured, and the average Transactions Per Second (TPS) was calculated using the ratio of completed transactions to total execution time.

Gas consumption statistics were extracted from the transaction effects generated by the Sui Testnet. These values include computation cost, storage cost, and storage rebate, providing insight into the computational and storage requirements of the implemented marketplace operations.

5.3 Latency Analysis

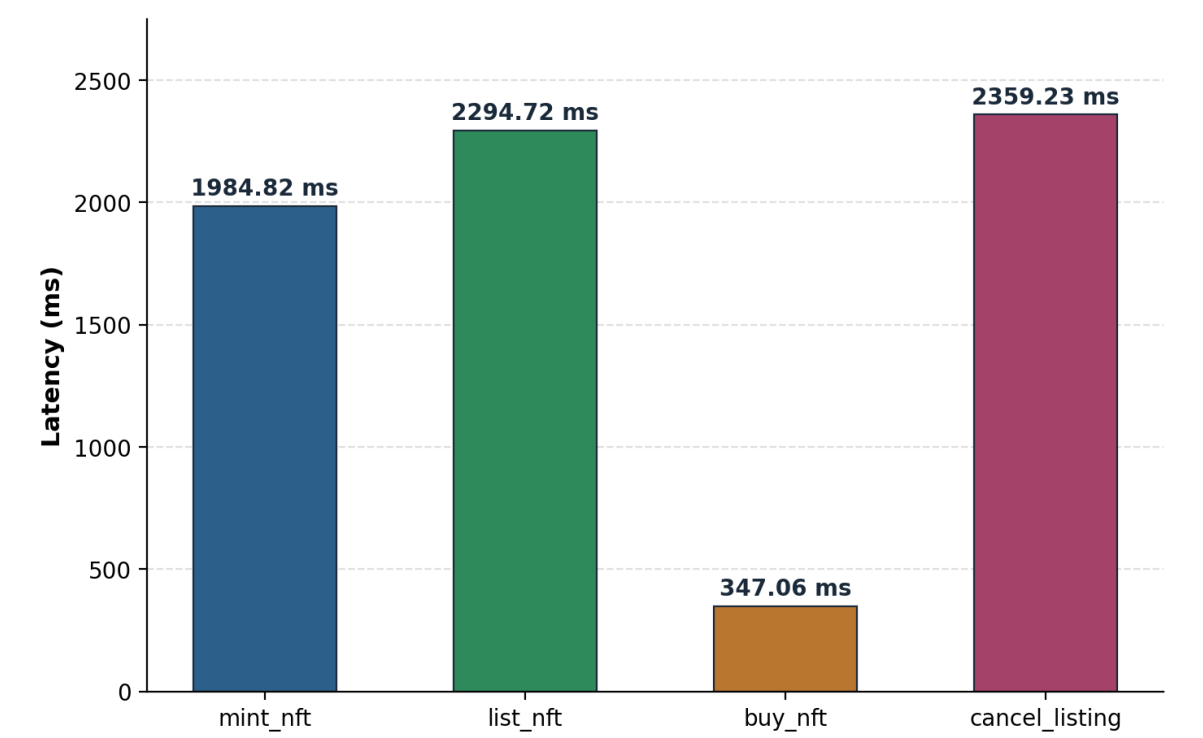
Transaction latency represents the time required for a blockchain network to process and confirm a transaction. Lower latency improves user experience by reducing the waiting time between transaction submission and completion.

Table 5.1 presents the measured latency for each marketplace operation.

Table 5.1 Transaction Latency

Operation	Latency (ms)
mint_nft	1984.82
list_nft	2294.72
buy_nft	347.06
cancel_listing	2359.23

Figure 5.1 Transaction Latency Comparison



Discussion

The experimental results reveal noticeable differences in execution time across the four marketplace operations.

The **buy_nft** operation recorded the lowest latency of approximately **347 ms**, making it the fastest operation in the benchmark. This comparatively low execution time can be attributed to the fact that purchasing primarily involves transferring ownership of an existing NFT and updating marketplace records, requiring relatively little object creation or storage allocation.

The **mint_nft** operation required approximately **1.98 seconds** to complete. Minting involves creating a completely new blockchain object, initializing metadata, allocating storage, and assigning ownership to the creator. These additional operations naturally increase execution time compared to ownership transfers.

The **list_nft** operation exhibited a latency of approximately **2.29 seconds**, while **cancel_listing** required approximately **2.36 seconds**. Both operations interact with the shared marketplace object, which must remain synchronized across validators. Since modifications to shared objects require consensus, these operations experience higher latency than transactions involving only independently owned objects.

Overall, the measured latency reflects the architectural characteristics of the Sui blockchain rather than inefficiencies in the marketplace implementation. Operations involving shared state incur additional coordination overhead, whereas ownership transfers involving existing objects execute considerably faster.

5.4 Throughput Evaluation

Throughput measures the number of transactions that can be processed within a given period and is commonly expressed in **Transactions Per Second (TPS)**. While theoretical throughput values provide insight into the maximum capabilities of a blockchain, practical measurements obtained through real deployments offer a more realistic evaluation of application performance.

For this experiment, five sequential NFT minting transactions were executed using the Sui CLI.

The observed results are summarized below.

Table 5.2 Throughput Benchmark

Metric	Value
Batch Size	5 Transactions

Metric	Value
Total Execution Time	10.29 Seconds
Throughput	0.486 TPS

The throughput was calculated as:

```
[
TPS = 5/10.29= 0.486
]
```

Discussion

The measured throughput of **0.486 TPS** should be interpreted carefully. It does **not** represent the maximum throughput capability of the Sui blockchain. Instead, it reflects the practical performance of sequential command-line execution on the public testnet.

Several factors contributed to the observed throughput:

- Sequential transaction submission through the CLI.
- Network communication latency.
- Testnet validator workload.
- Confirmation delays associated with public infrastructure.

Sui is designed to support **parallel execution of independent transactions**, particularly those involving separate owned objects. Since the benchmark intentionally executed transactions one after another, the platform's parallel execution capabilities were not utilized.

Consequently, the measured TPS serves as a baseline for comparison rather than an indicator of Sui's theoretical scalability.

5.5 Gas Consumption Analysis

Gas represents the computational and storage resources consumed during transaction execution. Understanding gas usage is important because it directly influences transaction cost and overall application efficiency.

Table 5.3 summarizes the gas usage recorded for the evaluated marketplace operations.

Table 5.3 Gas Consumption

Operation Computation Cost Storage Cost Storage Rebate

mint_nft	1,000,000	3,192,000	978,120
list_nft	1,000,000	11,825,600	4,506,876

Discussion

The recorded gas data indicates that computation costs remained constant for both operations, while storage costs varied significantly. This observation suggests that the primary contributor to transaction cost is persistent storage rather than computational complexity.

The **mint_nft** operation incurs storage costs associated with creating a new NFT object and storing its metadata on-chain. Although object creation requires additional storage allocation, the overall storage requirement remains moderate because only a single object is created.

In contrast, the **list_nft** operation exhibits a substantially higher storage cost. Listing an NFT requires updating the shared marketplace object and storing additional listing information, resulting in increased on-chain storage usage. However, a significant portion of this cost is offset by the **storage rebate**, which refunds part of the storage fee when storage can be reclaimed in future transactions.

These observations demonstrate that storage operations constitute the dominant component of gas expenditure in the implemented marketplace.

5.6 Performance Discussion

The benchmark results demonstrate that the developed NFT marketplace performs reliably under public testnet conditions while accurately reflecting the architectural characteristics of the Sui blockchain.

The comparatively low latency of the purchase operation illustrates the efficiency of ownership transfers involving existing objects. Conversely, higher latency for listing and cancellation operations is expected because these transactions interact with shared marketplace state and therefore require validator consensus.

Similarly, the measured throughput should not be interpreted as the upper performance limit of the Sui blockchain. Since the evaluation relied on sequential CLI execution, it intentionally represents a conservative baseline suitable for reproducible academic benchmarking. Under production environments that exploit object-level parallelism and concurrent transaction submission, substantially higher throughput can be expected.

Chapter 6: Challenges, Optimization Strategies, and Conclusion

6.1 Challenges Encountered During Development

The development and deployment of the NFT marketplace on the Sui blockchain provided valuable practical experience with Move-based smart contract programming and decentralized application deployment. Although the final implementation successfully achieved all intended functionalities, several challenges were encountered throughout the development process.

One of the initial challenges involved understanding the **object-oriented programming model** of the Sui blockchain. Unlike traditional account-based blockchains, Sui represents every digital asset as an independent object. This required careful management of object ownership, object references, and object transfers during marketplace operations. Proper handling of owned and shared objects was particularly important to ensure that NFTs maintained correct ownership while allowing marketplace interactions.

Another challenge arose during the implementation of the **marketplace module**, where active NFT listings were maintained using a shared object. Since shared objects require consensus before modification, additional attention was required to correctly update marketplace records while preserving consistency across all users. Understanding the distinction between owned and shared objects was essential for implementing secure listing and cancellation operations.

Deployment on the public Sui Testnet also introduced practical considerations related to package publication, object identifiers, gas budgeting, and wallet management. Each deployment generated unique package and object identifiers that had to be referenced accurately during subsequent transactions. Careful management of these identifiers was necessary to ensure successful interaction with the deployed smart contracts.

Finally, performance benchmarking was influenced by the characteristics of the public testnet environment. Transaction confirmation times varied depending on network conditions and validator workload. Since benchmarking was performed through sequential CLI execution, additional latency was introduced by command execution and network communication, resulting in throughput values that represent practical baseline measurements rather than the maximum capability of the Sui blockchain.

6.2 Optimization Strategies

Although the developed marketplace successfully satisfies its functional requirements, several optimization opportunities could further improve its performance and scalability.

The most significant optimization involves exploiting **Sui's parallel transaction execution** capabilities. In the current evaluation, transactions were executed sequentially using the Sui CLI. In a production deployment, independent transactions involving different NFT objects could be submitted concurrently, allowing the blockchain to process them in parallel and substantially increasing overall throughput.

Another optimization strategy involves reducing contention on shared marketplace objects. Since operations such as listing and cancellation require modifications to shared state, partitioning marketplace data or introducing additional indexing mechanisms could reduce synchronization overhead and improve responsiveness under heavy workloads.

Performance benchmarking could also be extended beyond simple sequential execution by introducing concurrent clients, larger transaction batches, and long-duration stress testing. Such experiments would provide a more comprehensive understanding of the marketplace's behaviour under realistic production conditions.

From a smart contract perspective, future versions of the marketplace could incorporate additional features such as auction-based sales, royalty distribution, advanced search mechanisms, and enhanced loyalty reward systems. These extensions would increase the functionality of the marketplace while demonstrating the flexibility of Sui's object-oriented architecture.

Finally, integrating decentralized storage solutions such as the **InterPlanetary File System (IPFS)** for NFT metadata and media assets would improve scalability by reducing on-chain storage requirements while maintaining secure references to digital content.

6.3 Overall Evaluation

The implementation and evaluation presented in this report demonstrate that the Sui blockchain provides a robust platform for developing decentralized NFT marketplaces. Its object-centric architecture aligns naturally with the ownership requirements of NFTs, while the Move programming language offers strong guarantees regarding resource safety and asset integrity.

The successful deployment of the marketplace on the Sui Testnet confirmed that all implemented smart contract modules operated correctly under real blockchain conditions. Functional validation demonstrated the successful execution of NFT minting, listing, purchasing, and listing cancellation, with ownership remaining consistent throughout every stage of the marketplace lifecycle.

Performance benchmarking further showed that the marketplace delivers reliable transaction execution under testnet conditions. While operations involving shared marketplace objects exhibited higher latency due to consensus requirements, ownership

transfer operations executed considerably faster. The measured throughput reflects sequential execution and should therefore be interpreted as a practical baseline rather than the full performance potential of the Sui blockchain.

Overall, the developed marketplace effectively demonstrates the practical application of Sui's object-oriented execution model in supporting secure and scalable NFT trading.

6.4 Conclusion

This report presented the complete implementation, deployment, and performance evaluation of a decentralized NFT marketplace developed on the Sui blockchain. The marketplace was implemented using the Move programming language and organized into three modular smart contract components responsible for NFT management, marketplace operations, and loyalty tracking.

The smart contract package compiled successfully, passed all unit tests, and was deployed without errors on the Sui Testnet. Functional validation confirmed the successful execution of all core marketplace operations, including NFT minting, listing, purchasing, and listing cancellation. These results demonstrate the correctness and reliability of the implemented marketplace under real blockchain conditions.

Performance benchmarking provided practical measurements of transaction latency, throughput, and gas consumption. The results highlighted the efficiency of ownership transfer operations while illustrating the additional coordination required for transactions involving shared marketplace state. Although the observed throughput was limited by sequential CLI execution and testnet conditions, the underlying architecture of Sui offers significant opportunities for higher scalability through parallel transaction processing.

In conclusion, the developed NFT marketplace successfully meets the objectives of the project by providing a secure, functional, and extensible decentralized trading platform. The findings presented in this report indicate that Sui's object-centric architecture and Move-based programming model provide a strong foundation for NFT marketplace applications, while also offering considerable potential for future enhancements and large-scale deployment.